

OPINION

A new era of financial services: How AI enhances investment efficiency

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1 | INTRODUCTION

Financial investment is an important part of the modern economy, promoting economic growth and wealth accumulation through the efficient allocation of capital. However, with the rapid development of global financial markets, the investment environment has become increasingly complex. Investors not only need to cope with a large amount of data and information, but also to capture market opportunities and avoid risks in a timely manner. Traditional investment analysis methods and tools are often overwhelmed when dealing with these complexities.

Over the past period of time, the rapid advancement of artificial intelligence (AI) technology has brought new hope to financial investment (Holzinger et al., 2023). Through its powerful data processing capabilities, pattern recognition, and predictive analytics, AI is able to cope with the complexity and dynamics of the financial market, effectively enhancing the efficiency of traditional financial institutions and demonstrating great potential and broad application prospects.

Financial complex systems are networks of multiple interconnected financial entities and activities that exhibit complex interactions and dependencies. These systems typically exhibit nonlinear behavior, dynamic evolution, and have self-organizing features. Traders, financial firms, and investors, as the core elements of financial complex systems, together constitute the operating mechanism of financial investment markets through complex interactions and information exchange.

In this study, we will discuss how AI technology can empower financial investments (Ahmed et al., 2022) to enhance their efficiency from the perspective of financial complex systems and analyze their limitations and potential drawbacks from a new perspective. The rapid development and application of AI technology, especially in the sector of financial investment, not only foretells a fundamental change in the way the financial market operates, but also strengthens the technological foundation and clarifies the potential direction for the

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future development of the financial industry. Digital intelligence (Vijayakumar et al., 2022) finance will accelerate into a new era.

1.1 | AI simplifies complex financial trading activities

The financial market is an intricate and ever-evolving landscape where trading operations entail the processing of vast amounts of data, instantaneous analysis, and swift decision-making (Sharma & Gupta, 2024). Conventional trading practices, heavily dependent on extensive human resources and intricate computational models, are not only labor-intensive and time-consuming but also susceptible to delayed judgments and suboptimal decisions. As financial markets become increasingly globalized and technology races ahead at an unprecedented speed (Ahmed et al., 2022), the intricacy of trading operations grows, compelling traders to make swifter and more accurate decisions.

In this context, AI technology shows strong potential to simplify complex financial transaction activities and improve efficiency and accuracy (Tanaka et al., 2024). AI technology helps financial institutions and investors cope with the complexity and uncertainty of the market by judging market trends, optimizing portfolios and providing real-time decision support, thus effectively improving financial investment efficiency at all levels.

1.2 | Market trend prediction

For traders, review is a crucial part of the investment process. By analyzing technical indicators of the candlestick charts, the review process seeks out the characteristics of a stock's trend and forecasts the stock price movements. However, this process is not only time-consuming and labor-intensive but also demands that investors possess profound technical analysis expertise and sharp judgment.

AI technology showcases its capability to forecast price movements across a diverse array of assets, including stocks, bonds, commodities, and currencies, by analyzing various factors such as historical price data, trading volumes, market news, and socioeconomic indicators. Currently, the application of AI in image recognition has become highly advanced, especially with the development of convolutional neural network (CNN) technology (Liu et al., 2022).

Based on this idea, a study that uses CNNs for financial image analysis to forecast stock price trends was proposed. The method uses a CNN to directly extract features from historical market data implicit in candlestick charts, such as daily opening price, daily high, daily low, daily closing price, and trading volume, to model the relationship between the standardized charts of price and volume data and the future stock price movements, thus enabling the prediction of stock prices.

The authors selected US stock price data from 1993 to 2019 for prediction. The results demonstrate that the CNN-based approach significantly outperforms traditional trading strategies (e.g., price momentum and reversal) and one-dimensional time series forecasting methods (e.g., ARIMA and Long Short-Term Memory Networks [LSTM]). In predicting US stock returns, the CNN-based approach achieved an accuracy exceeding 53% and demonstrated exceptional performance in portfolio strategies, with an annualized Sharpe ratio of up to 2.4 for equal-weight portfolios. Additionally, through transfer learning, the CNN-based approach also performs remarkably well across different markets and time scales, providing a powerful tool for global investment strategies.

This method not only substantially improves the efficiency of reviewing the market but also reduces errors in human analysis and judgment, enabling investors to focus more on strategy development and risk management, and enhancing the accuracy of trading decisions.

1.3 | Portfolio optimization

In today's complex financial environment, portfolio optimization (Montazeri et al., 2023) is undoubtedly a highly challenging task. It requires investors to conduct detailed risk assessments of different asset classes under uncertain market conditions to identify potential opportunities and risks hidden within the market. This allows for dynamic optimization and adjustment of large-scale investment portfolios (Buonaiuto et al., 2023).

The application of AI technology in the realm of investment portfolio optimization assists investors in constructing more robust portfolios through the sophisticated analysis and fine-tuning of the risk-to-return ratio. AI not only extracts features such as mean, variance, and correlation from vast data sets like stock prices and economic indicators to better understand market dynamics and asset performance, but it also employs machine learning algorithms such as decision trees, random forests, support vector machines, and neural networks to build portfolio optimization models by assessing the potential risks and returns of various types of assets (Kazemi et al., 2022). These models undergo continuous learning and adaptation, thereby improving their predictive accuracy and responsiveness.

Recently, deep reinforcement learning has been interaged into the domain of asset portfolio optimization, utilizing the deep Q learning algorithm to inform asset trading decisions. In their study, each asset is treated as an environment, and the strategy is trained by considering the income from holding the asset as its return and the average income from other portfolio assets as the cash return. The reward is calculated based on the average return of the asset group, establishing a connection between the trading return in a single asset environment and the overall portfolio return. With a single hyperparameter setting, the trained strategy demonstrates adaptability to various trading scenarios, achieving optimal performance in 36 out of 48 experimental groups. Compared to those traditional mathematical analysis methods for estimating risk premiums or return predictions, AI technology can more effectively capturing profit opportunities under temporary market inefficiencies, thus enabling more accurate portfolio optimization strategies.

1.4 | Real-time decision support

With the rapid development of financial markets, real-time monitoring of unusual trading behavior has become critical. The complexity and high degree of interconnectedness of financial markets mean that any unusual trading activity can trigger a chain reaction, leading to market volatility and potential financial crisis. Unusual trading behaviors may include insider trading, market manipulation, money laundering, and other forms of financial fraud. These behaviors not only undermine the fairness and stability of markets, but can also cause significant losses to investors and financial institutions. Traditional monitoring methods, which rely on manual review and post-event analysis, are often unable to detect and respond to abnormal trading behaviors promptly, leading to regulatory lag and risk accumulation. This poses a significant challenge for regulatory authorities.

In this context, the application of AI technology provides a new solution for real-time detection and adjustment of abnormal trading behaviors. Utilizing machine learning and big data analysis (Based Multidimensional Big Data Analytics over Clouds via Multi-Columnar Big OLAP Data Cube Compression), AI can rapidly process and analyze vast quantities of trading data to pinpoint abnormal trading patterns and potential risk factors. The real-time monitoring system is able to analyze transactions as they occur and making immediate decisions, thereby detecting and alerting to abnormal behavior in a timely fashion.

Some researchers have proposed a real-time monitoring framework based on big data technology by integrating real-time trading data, financial news, social media content, and

macroeconomic indicators, and this study used efficient data stream processing technology (e.g., Apache Kafka and Apache Flink) to construct an efficient real-time data stream processing framework. Integrating machine learning algorithms such as Isolation Forest for anomaly detection, LSTM for time-series analysis, and Graph Neural Networks for network behavior analysis, the proposed algorithm combination is capable of effectively identifying complex anomalous trading patterns (Bebin & Misbha, 2022; Prakash et al., 2023). The real-time monitoring and response system is designed to dynamically adjust thresholds, trigger alerts instantly, and generate detailed reports on abnormal behavior. This provides regulatory authorities with enhanced capabilities for swiftly responding to market anomalies, significantly improving both the timeliness and accuracy of anomaly detection.

1.5 | AI reduce operating costs and expenses

In today's highly competitive and rapidly evolving financial industry, managing operational costs and expenses is crucial for maintaining profitability and competitive edge. Financial institutions encounter various operational challenges, including elevated labor costs, cumbersome routine tasks, and complex risk management and data analytics requirements. The extensive application of AI technology in the financial sector not only demonstrates its robust potential for automating processes, enhancing risk management, optimizing customer service, and refining data analytics but also offers significant cost advantages, enhancing quality and efficiency in the financial investment sector.

1.6 | Intelligent customer service system

Intelligent customer service system is one of the major applications of AI technology in the financial investment sector. Traditional customer service relies on a large number of human resources, which not only consumes a large amount of human resources costs, but also may have the risk of data loss due to the confusion of manual records.

An AI-driven intelligent customer service system integrates natural language processing (NLP), machine learning, and other AI technologies to automate the handling of customer service requests. These systems can understand and interpret user queries, provide instant and accurate responses, or direct users to appropriate solutions. Intelligent customer service reduces dependency on manual customer service, significantly lowers labor costs, and greatly improves response speed and customer satisfaction. This is due to its ability to operate 24/7 and manage a large volume of inquiries autonomously.

The “Gong Xiaozhi” intelligent customer service system, launched by the Industrial and Commercial Bank of China in 2016, leverages advanced NLP technologies to achieve a 98% recognition rate of customer needs, setting a high standard in the industry. This enables rapid comprehension and efficient resolution of user inquiries, offering instant responses and significantly boosting service efficiency. The system operates seamlessly across over 20 mainstream channels, accommodating the diverse consultation needs of users and enhancing service convenience and user satisfaction.

By integrating big data, customer profiling, and self-learning technologies, “Gong Xiaozhi” delivers comprehensive one-stop services, including proactive suggestions, query resolution, marketing recommendations, business transactions, and after-sales tracking. Additionally, it offers engaging conversational features, making interactions more enjoyable for customers. This system exemplifies the transformative impact of AI in financial customer services, enhancing operational efficiency and customer experience through innovative applications.

1.7 | Automated manual routine tasks

The financial sector requires a high level of precision and data protection, as even minor human errors can lead to significant risks. Recently, Robotic Process Automation (RPA) (Perdana et al., 2023) has seen rapid advancements both domestically and internationally. The integration of RPA with technologies such as NLP and computer vision has substantially enhanced its problem-solving capabilities. AI-powered RPA effectively automates repetitive tasks that follow well-defined rules, adhering to predefined regulations and procedures. These tasks, including data entry and report generation, are often tedious and monotonous when performed manually, resulting in potential errors and considerable time consumption.

Automating these routine tasks not only increases efficiency but also optimizes human resources by relieving employees from repetitive duties. This transition enables employees to focus on higher-value activities, such as strategic planning and innovative business development, thereby enhancing job satisfaction, fostering creativity, and improving overall operational efficiency and competitiveness.

In 2023, Hangzhou Bank introduced an intelligent robot equipped with RPA technology to automatically analyze and process loan applications, as well as handle other operational tasks. These robots are capable of mimicking the actions of human loan officers, automatically performing tasks such as processing loan applications, verifying applicant details, assessing credit risks, and making approval decisions. This automation streamlines the loan approval process, making it more efficient, accurate, and reliable, which in turn reduces approval times and boosts customer satisfaction. Moreover, the smart robots can autonomously collect applicants’ personal data, financial standing, and credit histories from diverse data sources. Utilizing optical character recognition and NLP technologies, they can verify the authenticity and accuracy of the documents and information submitted by applicants. These “digital employees” can operate 24/7, significantly reducing the processing time for individual business tasks, and achieve a 100% accuracy rate (Mihalcea et al., 2006) (Figure 1).

In conclusion, through automation driven by AI and machine learning, financial service providers can significantly reduce costs and deliver more personalized and efficient services to meet the growing demands of modern consumers. With the support of Artificial Narrow Intelligence (ANI), the efficiency of financial institutions in handling individual tasks has markedly improved (Holzinger et al., 2023). As digital systems evolve and large model technologies are applied, it is expected that the accuracy and efficiency of these services will further expand, ushering the financial industry into a new era of heightened efficiency and security.

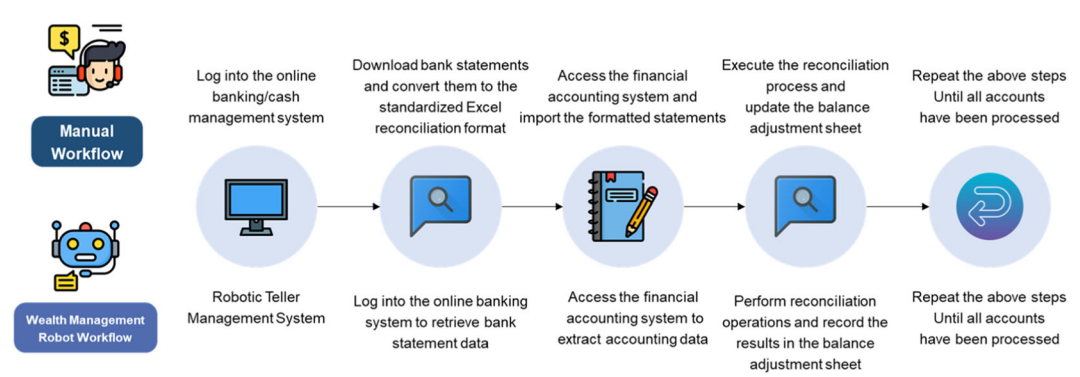


FIGURE 1 Comparison between manual bank reconciliation process and financial robot workflow.

1.8 | AI enhances financial inclusion

Since the beginning of 2023, the emergence of large model technologies, exemplified by ChatGPT with its powerful content generation and text comprehension capabilities, has sparked a new wave of AI innovation worldwide. The financial industry, with its rich resources in information, data, knowledge, and talent, offers an ideal environment for the exploration and implementation of large model applications. Leveraging advanced digital infrastructure, large model technology is accelerating its deep integration with the financial sector. This integration is expanding the boundaries of digital financial technology and applications, rapidly and profoundly transforming the financial service experience. It provides customers with more efficient and personalized financial services, continuously enhancing the efficiency of the financial system.

1.9 | Large models empowered Robo-Advisor

A Robo-Advisor is a financial service that employs advanced financial models and algorithmic technology to offer automated financial planning and investment management to investors. With the assistance of ANI, although banks' smart customer service systems have been able to efficiently handle simple tasks, they still suffer from several pain points, including high knowledge maintenance requirements, low question-and-answer coverage, and so on. Due to restricted knowledge boundaries, the quality of responses to out-of-scope questions is often poor, leading to potential risks for customer experience and the company's image (Wang et al., 2024).

The emergence and rapid development of large models have greatly improved the accuracy of market analysis and broadened the knowledge of intelligent customer service, providing personalized services for various types of users and enhancing the effectiveness and reliability of intelligent customer service scenarios.

In the face of a rapidly changing market environment, nonprofessional investors often struggle to access actionable information and miss optimal investment opportunities. Large models can serve as the "brain" for information analysis, integrating with various external APIs to obtain the latest financial data, market data, and other relevant information. They assess potential investment opportunities and guide users towards making informed decisions.

Morgan Stanley has significantly enhanced its wealth management division by integrating OpenAI's general large model GPT-4. This advanced AI assistant enables financial advisors to quickly access and synthesize vast amounts of internal data, streamlining the retrieval of investment strategies, market analyses, and research papers (Menghao et al., 2023). By leveraging the collective expertise of the firm's top analysts and strategists, the advanced AI assistant can provide timely and well-informed advice to various kinds of clients. This transformation not only improves decision-making speed but also elevates the overall client service experience, showcasing the substantial advantages GPT-4 brings to the financial sector.

Similarly, JPMorgan Chase has developed a "Fed-whispering" AI model, utilizing advanced NLP technology to analyze the public speeches and statements of Federal Reserve officials. This AI tool, based on a large language model similar to ChatGPT, examines 25 years of Fed communications to gauge their policy stance, effectively assigning a "Hawk-Dove" score. This score indicates the likelihood of policy changes, such as interest rate hikes (hawkish) or cuts (dovish). The AI model has demonstrated its value in predicting policy shifts and providing actionable trading signals, thereby assisting investors in making informed decisions based on anticipated Federal Reserve policies (Biswas, 2023).

1.10 | Large models improve financial risk management efficiency

Effective financial risk management is a cornerstone for financial institutions to ensure operational stability, mitigate losses, and protect client assets. A robust risk management framework not only shields institutions from the vicissitudes of the market, credit exposure, and operational hazards but also strengthens their competitive edge and market credibility. As financial markets grow increasingly intricate and unpredictable, conventional risk management approaches are revealing limitations in terms of efficiency and responsiveness. The advent and adoption of large-scale model technology have introduced a new paradigm and transformative potential to the field of financial risk management.

Tencent Cloud's financial risk control large model provides financial institutions with a more efficient and intelligent risk management solution. This model is embedded with extensive knowledge of illicit activities and fraud prevention, enabling it to automatically achieve expert-level precision in risk control. Even without specialized modeling talent, financial institutions can efficiently construct risk control models using this technology. Additionally, the model excels in addressing the challenges of “small sample” and even “zero sample” scenarios. By constructing an intelligent investment scenario simulation system and integrating financial data and market information, the financial risk control large model can simulate various complex market scenarios such as economic downturns, market crashes, or industry fluctuations. For instance, Dongfeng Nissan Auto Finance successfully utilized Tencent Cloud's risk control model to create customized risk control strategies despite having limited samples. This approach significantly reduced modeling time and improved antifraud effectiveness by ~20% in terms of Kolmogorov–Smirnov scores compared to traditional methods (Figure 2).

The implementation and application of financial large models empower financial participants by significantly enhancing the scope and depth of financial information collection, organization, and analysis. This greatly reduces information asymmetry, lowers transaction costs, and effectively boosts the efficiency of financial investments. On the one hand, financial institutions need to provide financial product services in the financial market, and the application of large models helps financial institutions to enhance the efficiency of risk

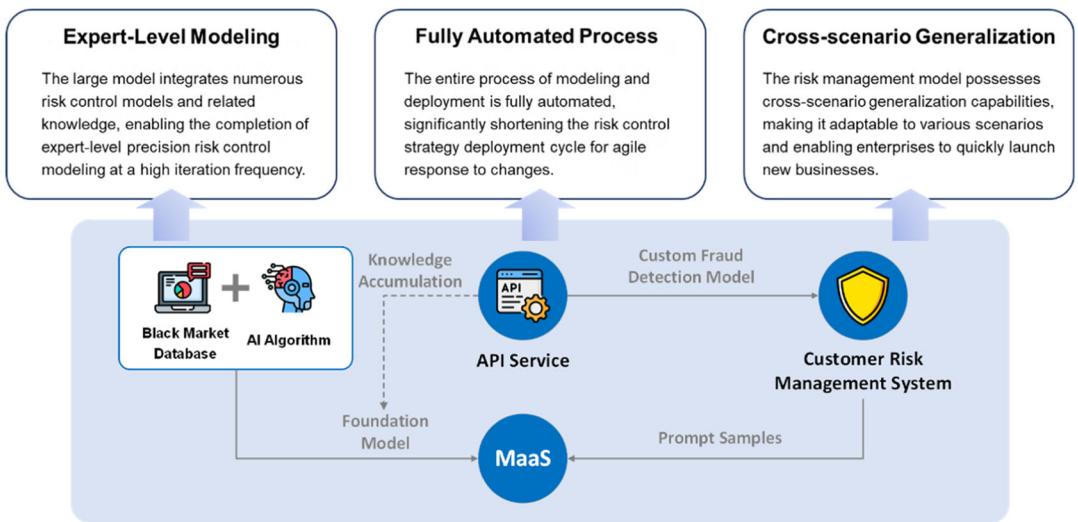


FIGURE 2 Customized dynamic risk management system based on Tencent Cloud financial risk control large model.

management, improve the quality of financial services, and reduce the cost of financial services by promoting the reduction of operating costs. This not only brings higher profit margins for financial institutions, but also can further optimize the allocation of resources, improve the efficiency of capital utilization, and promote the stable and healthy development of the financial market. On the other hand, the majority of investors need to obtain financial product services from the financial market and the application of large models, such as intelligent investment advisors and intelligent marketing, can help investors reduce the cost of information acquisition, enhance access to information, assist in optimizing the quality of financial decision-making, and force the main body of the financial market to strengthen information transparency and operational efficiency, further promoting the healthy development of the financial industry.

1.11 | Challenges and obstacles in implementing financial large models

Over the past year, numerous financial enterprises have begun experimenting with large models, leading to a proliferation of various general-purpose and finance-specific large models. The application range and business scenarios for these models are rapidly expanding. However, a series of potential risks and drawbacks are also becoming apparent, somewhat limiting the potential of AI large models in the financial investment sector.

Fueled by market frenzy and policy guidance, nearly every financial institution is attempting to introduce large models despite their incomplete maturity. The technological immaturity often results in large models being unable to accurately calculate and analyze the return on investment of financial investments, making it difficult to precisely quantify their business value and lacking an interpretable economic significance (Balakir et al., 2020).

Furthermore, the current large model technology has yet to deeply integrate with financial investment operations. The financial expertise and contextual understanding of these models are generally inadequate. This results in existing large models struggling to deeply involve in complex reasoning, real-time trading, and other sophisticated scenarios.

Efficient operation of financial large models requires not only extensive data sets from the financial sector but also a clear comprehension of specific financial terminologies and commands to better grasp complex financial contexts and events. For these models to be truly effective in the financial domain, they need to undergo substantial business customization, incorporating scientific trading data, financial jargon, trading methodologies, business rules, and even regulatory frameworks. Achieving this level of integration and enhancing model accuracy from around 70% to over 95% is not an easy process. It involves extensive data processing and cleaning, and for enterprises, it necessitates embedding substantial business knowledge and professional expertise into the system. This poses quite high demands on the knowledge capacity and business acumen of the engineers within the enterprise.

Finally, it is crucial for large models to address the issue of “hallucinations” to improve their accuracy and interpretability. When the training data contains inaccuracies, biases, or fictional content, or when there is inherent ambiguity or uncertainty in the input, large models may misinterpret market dynamics, economic indicators, or risk factors, leading to outputs that appear reasonable but are actually baseless or incorrect (Shukla et al., 2023). Such misleading outputs can cause financial institutions to make unsound asset allocations, incorrect credit decisions, or erroneous market trend predictions, potentially resulting in economic losses, reputational damage, and even systemic financial risks (Figure 3).

The financial sector's demand for high accuracy and interpretability necessitates effective handling of hallucination issues to reduce the risk of misleading decisions (Herman et al., 2023). Therefore, enterprises need to integrate diverse data and knowledge resources, including proprietary financial data and knowledge graphs, to guide the model towards correct

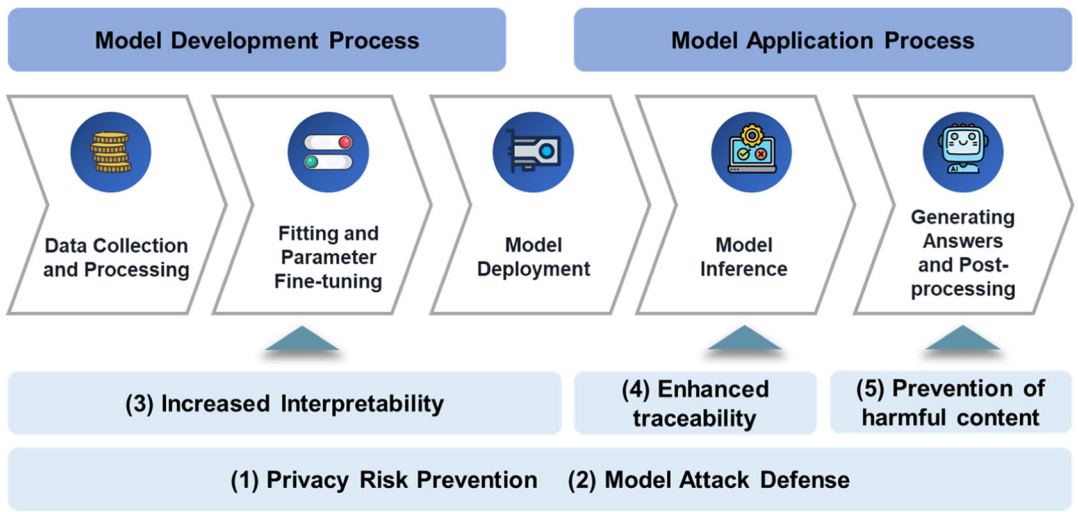


FIGURE 3 Risk prevention and control in the development framework of large models.

and precise understanding of the financial industry, thereby enhancing model interpretability. Investors also need to improve their ability to discern information, recognizing the dangers of hallucination issues to avoid unnecessary financial losses.

2 | CONCLUSION

The wide application of AI in financial investment has significantly enhanced the efficiency of interconnected financial entities and markets within the financial ecosystem, injecting new vitality into the financial sector. For traders, AI technology aids in trend prediction, portfolio optimization, and real-time decision-making, greatly simplifying complex trading activities in an information-intensive era. For financial institutions, AI-driven intelligent customer service systems and RPA effectively enhance service efficiency while substantially reducing operational costs. For investors, large models improve the ability to collect and analyze financial information and data, thereby enhancing the quality of participation and decision-making in financial investments.

AI has injected boundless vitality into financial technology, significantly enhancing the efficiency of financial investments, optimizing industry services, and increasingly emerging as a key force for future changes in the financial industry. Looking ahead, as AI technology continues to advance and innovate, its application in financial investments will become more extensive and profound. As integral components of the financial complex system, traders, financial institutions, investors, and financial market regulators must collaborate closely to address challenges such as over-reliance on AI, algorithmic spoofing, model hallucinations, and legal and ethical risks. By exploring solutions to these risks, they can promote the healthy and sustainable development of the financial investment sector and usher financial services into a new era.

AUTHOR CONTRIBUTIONS

Zhiyi Liu: Conceptualization; data analysis; writing—original draft. **Kai Zhang:** Data collection; review of literature; data analysis. **Hongyi Zhang:** Supervision; coding.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data used in this study were sourced from publicly available data sets found on the internet. The specific URLs or data repositories where the data were obtained are provided in the methodology section of the paper. The data are freely accessible and do not require any special permissions or licenses for use.

ETHICS STATEMENT

Not applicable.

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REFERENCES

- Ahmed, Imran, Gwanggil Jeon, and Francesco Piccialli. 2022. "From Artificial Intelligence to Explainable Artificial Intelligence in Industry 4.0: A Survey on What, How, and Where." *IEEE Transactions on Industrial Informatics* 18(8): 5031–42. <https://doi.org/10.1109/TII.2022.3146552>
- Balakir, Artsiom, Alan Yang, and Elyse Rosenbaum. 2020. "An Interpretable Predictive Model for Early Detection of Hardware Failure." In *2020 IEEE International Reliability Physics Symposium (IRPS)*, 1–5. <https://doi.org/10.1109/IRPS45951.2020.9129615>
- Bebin, Josey T., and D. S. Misbha. 2022. "Detection of Attacks Using Attention-Based Conv-LSTM and Bi-LSTM in Industrial Internet of Things." In *2022 International Conference on Automation, Computing and Renewable Systems (ICACRS)*, 402–7. <https://doi.org/10.1109/ICACRS55517.2022.10029012>
- Biswas, Som. 2023. "Importance of Chat GPT in Agriculture: According to Chat GPT." SSRN Scholarly Paper. Rochester, NY. <https://doi.org/10.2139/ssrn.4405391>
- Buonaiuto, Giuseppe, Francesco Gargiulo, Giuseppe De Pietro, Massimo Esposito, and Marco Pota. 2023. "Best Practices for Portfolio Optimization By Quantum Computing, Experimented on Real Quantum Devices." *Scientific Reports* 13(1): 19434. <https://doi.org/10.1038/s41598-023-45392-w>
- Herman, Dylan, Cody Googin, Xiaoyuan Liu, Yue Sun, Alexey Galda, Ilya Safro, Marco Pistoia, and Yuri Alexeev. 2023. "Quantum Computing for Finance." *Nature Reviews Physics* 5(8): 450–65. <https://doi.org/10.1038/s42254-023-00603-1>
- Holzinger, Andreas, Katharina Keiblinger, Petr Holub, Kurt Zatloukal, and Heimo Müller. 2023. "AI for Life: Trends in Artificial Intelligence for Biotechnology." *New Biotechnology* 74(May): 16–24. <https://doi.org/10.1016/j.nbt.2023.02.001>
- Kazemi, Alireza, Reza Boostani, Mahmoud Odeh, and Mohammad Rasmi AL-Mousa. 2022. "Two-Layer SVM, Towards Deep Statistical Learning." In *2022 International Engineering Conference on Electrical, Energy, and Artificial Intelligence (EICEEAI)*, 1–6. <https://doi.org/10.1109/EICEEAI56378.2022.10050469>
- Liu, Hong, Jing Wang, Shui Jiang, and Xiaohua Song. 2022. "Research on Fixed Asset Management from the Perspective of Industry-Finance Integration—Based on CNN Method." In *2022 3rd International Conference on Education, Knowledge and Information Management (ICEKIM)*, 748–54. <https://doi.org/10.1109/ICEKIM55072.2022.00166>
- Menghao, Li Minggao, Yi Fengchao, Li Yagang, and ShiYajing. 2023. "Research on the Application of LLM in Power Finance Middle Platform." In *2023 2nd Asian Conference on Frontiers of Power and Energy (ACFPE)*, 282–9. <https://doi.org/10.1109/ACFPE59335.2023.10455549>
- Mihalcea, Rada, Hugo Liu, and Henry Lieberman. 2006. "NLP (Natural Language Processing) for NLP (Natural Language Programming)." In *Computational Linguistics and Intelligent Text Processing*, edited by Alexander Gelbukh, 319–30. Berlin, Heidelberg: Springer. https://doi.org/10.1007/11671299_34
- Montazeri, Sina, Akram Mirzaeinia, Haseebullah Jumakhan, and Amir Mirzaeinia. 2023. "CNN-DRL for Scalable Actions in Finance." In *2023 International Conference on Computational Science and Computational Intelligence (CSCI)*, 302–6. <https://doi.org/10.1109/CSCI62032.2023.00053>
- Perdana, Arif, W. Eric Lee, and Chu Mui Kim. 2023. "Prototyping and Implementing Robotic Process Automation in Accounting Firms: Benefits, Challenges and Opportunities to Audit Automation." *International Journal of Accounting Information Systems* 51(December): 100641. <https://doi.org/10.1016/j.accinf.2023.100641>

- Prakash, Satya, Anand S. Jalal, and Pooja Pathak. 2023. "Forecasting COVID-19 Pandemic Using Prophet, LSTM, Hybrid GRU-LSTM, CNN-LSTM, Bi-LSTM and Stacked-LSTM for India." In *2023 6th International Conference on Information Systems and Computer Networks (ISCON)*, 1–6. <https://doi.org/10.1109/ISCON57294.2023.10112065>
- Sharma, Rishabh, and Shikhar Gupta. 2024. "Strategic Deployment of Deep Learning Algorithms to Mitigate Fraud in Online Finance." In *2024 7th International Conference on Circuit Power and Computing Technologies (ICCPCT)*, 1:1007–11. <https://doi.org/10.1109/ICCPCT61902.2024.10673115>
- Shukla, Neelesh K., Raghu Katikeri, Msp Raja, Gowtham Sivam, Shlok Yadav, Amit Vaid, and Shreenivas Prabhakararao. 2023. "Investigating Large Language Models for Financial Causality Detection in Multilingual Setup." In *2023 IEEE International Conference on Big Data (BigData)*, 2866–71. <https://doi.org/10.1109/BigData59044.2023.10386558>
- Tanaka, Hiroshi, Masaru Ide, Jun Yajima, Sachiko Onodera, Kazuki Munakata, and Nobukazu Yoshioka. 2024. "Taxonomy of Generative AI Applications for Risk Assessment." In *2024 IEEE/ACM 3rd International Conference on AI Engineering – Software Engineering for AI (CAIN)*, 288–9. <https://ieeexplore.ieee.org/document/10556211>
- Vijayakumar, Bharati S., Dhanya Pramod, and Ramakrishnan Raman. 2022. "Intention to Use Artificial Intelligence Services in Financial Investment Decisions." In *2022 International Conference on Decision Aid Sciences and Applications (DASA)*, 560–6. <https://doi.org/10.1109/DASA54658.2022.9765183>
- Wang, Lei, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, et al. 2024. "A Survey on Large Language Model Based Autonomous Agents." *Frontiers of Computer Science* 18(6): 186345. <https://doi.org/10.1007/s11704-024-40231-1>

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